INDICATION CALCULATOR · CONTROLLING & FINANCE · PRE-CONFERENCE DATEN-WG 2026

Visual standards calculator: Paid, Flexible or MacGyver?

What does it cost to implement visual standards in Power BI with **3rd-party visuals**, **open source**, **Deneb** or **core visuals**? The guiding principle: **standard first, only as bespoke as necessary.** So the calculator first checks whether there is a professional reason to depart from the core visuals, and then compares across three levels: Overview with three inputs, Narrow down via requirements and red flags, Details with every assumption and source. It gives an **indication** with ranges, not a price quote: every assumption is editable and marked with its evidence. **If this is your first time here, answer seven short questions in the guided start** — it only sets fields that you will find again, every one of them, in levels 2 and 3.

LEVEL 1 What does a standard cost you, per viewer per year?

Three inputs, everything else sits at its default: a five-year horizon, volume tiering (the more licences, the cheaper each one) and a mid-range chart mix. In each approach's bar the band shows the P10 to P90 range, and the line inside it shows **P50 — the middle number: in half of the simulation runs the cost falls below it**. Levels 2 and 3 get more precise.

Presets: Pilot · 1 report author / 10 viewers | Mid-market · 3 / 40 | Corporate · 25 / 1,500 | Corporate with Report Server · 15 / 400 | Large corporate · 200 / 10,000

Scenarios: Enterprise with Org Store · 25 / 1,500 | Public administration · 6 / 300 | **Finance with mandatory IBCS · 5 / 150** | Replacing an installed base · 8 / 400 | Build instead of buy · 1 / 10

Finance with mandatory IBCS · 5 / 150: The finance department the standard is made for: mostly management reporting, every report to the standard. So with Paid every reader needs a licence, and class A disappears. Not suited if only part of the reports need the standard — adjust the report classes instead.

Report authors: 5

People who build reports with the standard visuals.

Viewers (read-only): 150

With paid visuals, only the share of viewers who open class B or C reports needs a licence (default 85%, slider in level 3; with mandatory IBCS, all of them).

Operating model

Enterprise BI | **Managed Self-Service** | Self-Service

A core team maintains templates, trained report authors use them.

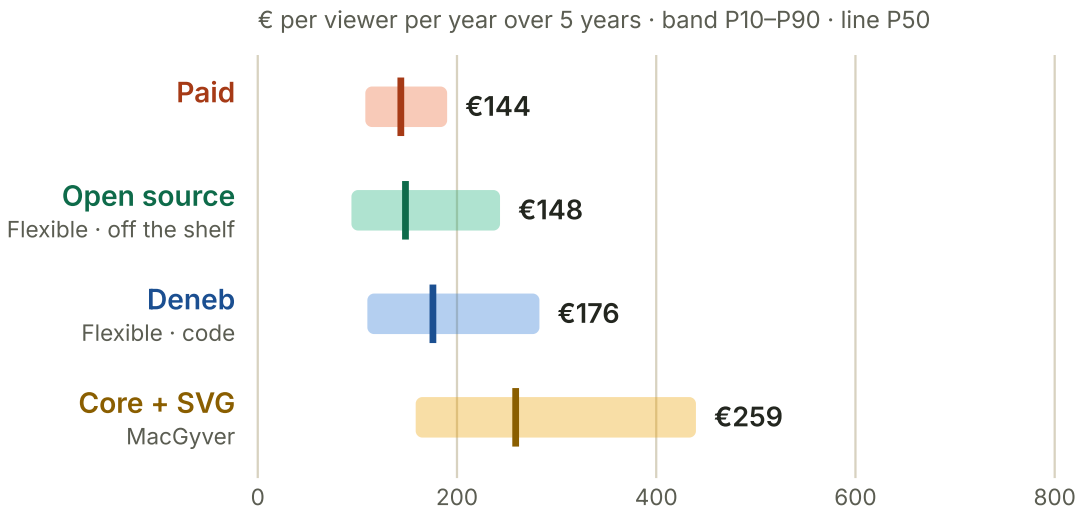
Price anchor for paid

Inforiver ~€33 | **Blended price ~€80** | Zebra level ~€170 (derived)

Per **licensed** user per year before volume tiering (licensed headcount: all report authors plus the share of viewers with access to standard reports). It flips the ranking more than any other assumption; the sentence on the right shows all three.

☐ A ready-made template library for Deneb and core is available
e.g. IBCS Chart Builder (the Daten-WG's Deneb templates, our own project), vegaviz, PowerofBI UDFs for SVG. Cuts initial build and reuse to 60 to 70%, training by half as much. Off by default, so that our own project does not quietly count in its own favour.

With 5 report authors and 150 viewers, **Paid and Open source** are level at around €144 per viewer per year (P50); Deneb costs 1.2 times as much, Core + SVG costs 1.8 times as much. From around **176 viewers** the licence per head becomes more expensive than the cheapest hours-based approach. Price anchor: Blended price €80 · without a template library · hours anchor: Panel (default). **13%** of the Paid cost rests on evidenced prices, **43%** on estimates (the rest: rules of thumb and the price anchor). The bar per approach is in level 2.



WHY THIS RESULT, AND WHAT TURNS IT

1 · Ratio 1 : 30. Paid pays €58,000 in licences over 5 years. Open source saves that, but needs 528 more hours, valued at €66,000. The two weigh about the same here, so the result is a tie.

2 · Licence price. Anchor €80 per licensed user per year (list price before tiering). Parity with Open source at around **€92**. That is the negotiation ceiling, not the list price.

3 · Hours per chart. Panel values per reused chart: paid 1 h, open source 2 h, Deneb 2.5 h, Core 4 h (unmeasured). Open source would have to cut its hours by around **6%** to draw level; the switch "template library" cuts Deneb and Core by 30 to 40%.

What holds is the direction of these three levers, not the second digit. The number of viewers, the licence price and the hours per chart explain the gap almost completely; everything else shifts it by percentage points. Key statements for discussion

What holds: the order of magnitude per viewer and the number of viewers from which the licence dominates (structural). **What does not hold:** the order of Paid and Open source, which hangs on unmeasured hour assumptions; amounts to more than one digit, and the benefit side. Provenance of the 58 assumptions: 3 verified, 9 search snippets, 3 rules of thumb, 9 simulated panel, 34 authors' estimate (level 3).
Who is ahead, depending on the price anchor (Inforiver / blended price / Zebra): without a template library Paid / Paid / Open source, with a template library Paid / Deneb / Deneb. Expected values, pure cost. If you have a real quote, enter it in level 2 and it replaces the anchor.

Get more precise: requirements, red flags, horizon → All details

If you want to know first **what drives this number**: open the map of the model → (level 2 — the click switches you there).

LEVEL 2 What drives the cost

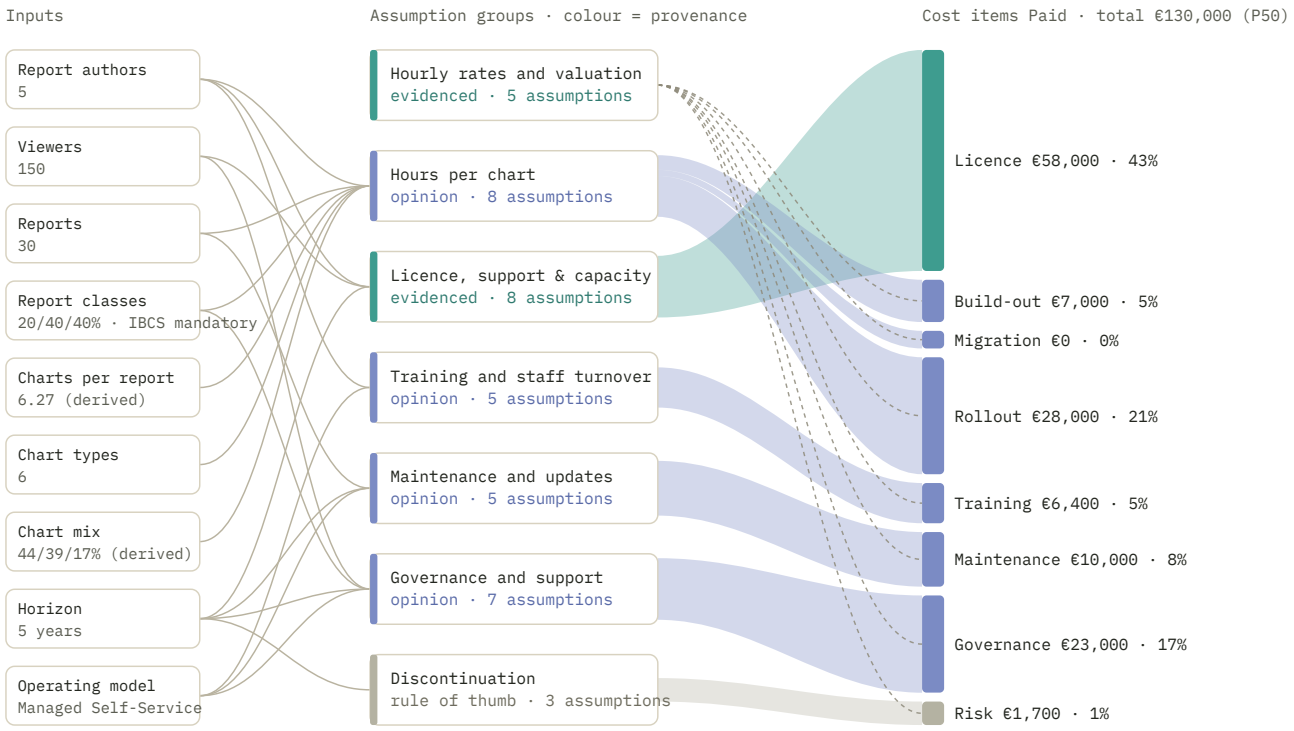
The map of the model: which quantities there are, where they come from and where they run to — and **every element is clickable, so you can change it right here**. The overview first, then the input sections one by one.

Impact map

BAND THICKNESS = SHARE OF P50

In short: **inputs → assumption groups → cost items** — on the left what you set, in the middle the assumptions with their provenance (colour as in the evidence bar), on the right the cost items of the selected approach. Clicking an element highlights its paths, explains it underneath and opens a field for changing it.

Paid | Open source | Deneb | Core + SVG



3rd-party paid: this approach is dominated by **Licence, support & capacity** at €58,000, which is 43% of the P50 cost of €130,000. That item is driven by the group "Licence, support & capacity", whose 8 assumptions are predominantly **evidenced** (5 evidenced, 0 rule of thumb, 3 panel or authors' estimate).

Clicking an element in the map — an input on the left, an assumption group in the middle or a cost item on the right — shows here **where it feeds in and how much it accounts for**, highlights the connected paths and opens a field next to it for changing the value. With the keyboard: tab to the element, then Enter or the space bar. A second click or Escape clears the selection.

evidenced = price list or snippet rule of thumb = literature without measurement

opinion = panel or authors' estimate

In detail: how to read the bands, lines and colours

Your inputs on the left, in the middle the assumption groups with their provenance (colour as in the evidence bar), on the right the cost items of the selected approach. Thick bands carry amounts and add up to the P50; thin lines only show which input feeds which group; dashed lines scale something without carrying an amount of their own. Amounts and shares appear on hover.

The tabs above the map switch the approach (paid, open source, Deneb, core + SVG); the map always shows the P50 cost of the selected approach. Green = evidenced (price list or snippet), grey = rule of thumb, blue = opinion — the same split as in the evidence bar in the result.

LEVEL 2

Volume assumptions

Who builds, who reads, how much gets built? Paid visuals scale with the **viewers who need a licence** (since v0.18 no longer with all of them, but with the share who open standard reports), every other approach scales with **chart types and report authors**. This ratio decides more than any list price.

Report authors **5**

People who build and maintain reports with the standard visuals.

Viewers (users who read reports) **150**

Microsoft scenario: around 9 viewers per report author. With paid visuals not every viewer needs a licence, only the share with access to standard reports (assumption “share of viewers with access to standard reports”, default 85%); governance, user support and reader time run across all viewers.

Viewer ceiling (cap) **0**

0 = no cap. Viewer growth (step 4) is capped at this value each year: viewers cannot grow beyond the workforce. Suggestion: your workforce, or failing that three times today's viewers. Affects licence, support, governance and demand.

Standard reports **30**

Reports that are built to the standard. Typical value for Managed Self-Service with 5 report authors: **20** (Enterprise ≈ 6, Managed ≈ 4, Self-Service ≈ 2 per report author, plus 2 per 1,000 viewers; from Daten-WG experience) apply

Charts per report calculated **6.27**

Derived from the report classes: share per class × charts per report of that class. The classes themselves can be changed further down.

Chart types in the kit **6**

Variance chart, waterfall, table with Δ columns, KPI card, time series AC/PY/PL/FC.

Chart mix by complexity (%) calculated **100%**

simple	44
medium	39
complex	17

Derived from the report classes (chart mix per class in level 3, weighted by share × charts per report). **Simple:** bars and columns with variances, lines, KPI cards. **Medium:** tables with embedded charts, small multiples, scenario notation AC/PY/PL/FC. **Complex:** waterfall with variances, combined charts, commentary, drilldown. The panel hours apply to a 50 / 35 / 15 mix; other mixes scale the hours per approach with the complexity factors from step 4.

Time horizon

3 years | **5 years** | 10 years

Forrester works with 3 years, Gartner TCO with 5. Ten years show how licence and maintenance overtake the upfront investment.

Report classes: who actually needs an IBCS chart? **100%**

CLASS	SHARE %	CHARTS/REPORT
A · Operational, no IBCS need Lists, tables, status overviews, simple bars — built with core visuals in every approach	20	8.36
B · IBCS standard Variance charts, time series with scenario notation AC/PY/PL/FC, tables with embedded charts	40	6.27
C · IBCS advanced Waterfall with variances, small multiples, commentary, drilldown	40	5.22

IBCS is mandatory: the 20% of class A reports are built with the selected approach as well, and every report needs a conformance review. That gives **6.3 charts per report** and the chart mix **44 / 39 / 17%**. **The obligation raises the licensed headcount to all viewers:** every report runs through the selected approach, so with 3rd-party paid every reader needs a licence — 150 viewers plus 5 report authors.

☑ **IBCS mandatory for all reports**
In the enterprise this is the real cost driver: once only IBCS-compliant reports are allowed, the operational class A reports get built with the selected approach as well — and every report needs a conformance review, equally expensive in every approach. Without this switch only classes B and C are reviewed.

Operating model

Enterprise BI | **Managed Self-Service** | Self-Service

Self-service share of report authors (hybrid) 25%

Managed Self-Service: A core team maintains templates, trained report authors use them. Here 2 of 5 report authors the templates.

Current approach (existing estate)

Greenfield, nothing in place

If you stay with today's approach, the switch costs €0. Every other approach carries the rebuild of the existing estate.

Existing reports that would have to be migrated 0

Reports on today's approach that would be rebuilt on a switch.

Four fine adjustments have moved to level 3: semantic models, adoption rate (standard compliance), uncertainty in the volume assumptions and the hours anchor. They are not gone, just one level deeper — their defaults still count in unchanged.

LEVEL 2

Vendor quotes

Otherwise the calculator works with **price anchors** from search snippets. If you have a real quote, you hold the hardest evidence in the whole model — so use it. An active quote replaces the anchor for "3rd-party paid": price, minimum term, price escalation clause and support then come from the quote, and volume tiering stays off.

Enter quotes

UP TO THREE · EXACTLY ONE CAN BE ACTIVE

A quote is commercially sensitive: vendor, price, term and note appear in plain text in the share link and in the Excel export — whoever gets the link reads your terms along with it. Without an active quote everything stays as before, and the three price anchors in level 1 still apply.

No quote recorded yet. The calculation runs on the price anchor from level 1 — a search snippet, not a commitment from any vendor.

+ Add quote

LEVEL 2

Rate the requirements

27 criteria. Prioritised by MoSCoW — must (cannot do without it), should (important), could (nice to have), won't (not this time). On a 1-to-5 scale the boxes show how well each approach meets the requirement. **Must + 1 = knock-out** (knock-out = the approach is excluded but stays visible in the result with its cost, so that the price of the rule can be seen), must + 2 = only via a workaround. Should and could feed into the requirements score. **Clicking a box overrides the rating** with your own assessment.

Requirements and fulfilment level

+ Own requirement | ↺ Reset ratings

REQUIREMENT	PRIORITY	PAID 3rd-party paid	FLEXIBLE · OFF THE SHELF Open source / in-house	FLEXIBLE · CODE Deneb / Vega
IBCS-compliant notation AC/PY/PL/FC scenarios, variances, scaling	**M** S C W	5	4	4
IBCS-certified visual Official certification of the software product	M S **C** W	5	1	1
Microsoft-certified visual Export, subscriptions, tenant approval without exception	M **S** C W	5	2	4
PDF/PPT export and e-mail subscriptions Only with certified visuals	M **S** C W	5	2	3
Data export from the visual (Excel/CSV) “The numbers behind the chart” for the recipient, including the tenant and report setting	M **S** C W	4	3	2
Interactivity Selection, cross-highlighting, standard tooltip	M **S** C W	5	4	4
Drilldown, drillthrough and report page tooltip Drill from region down to	M **S** C W	4	3	2

REQUIREMENT	PRIORITY	PAID 3rd-party paid	FLEXIBLE · OFF THE SHELF Open source / in-house	FLEXIBLE · CODE Deneb / Vega
country, open a detail page, show a report page as a tooltip				
Large data volumes per visual More than 30,000 rows in the visual's data window	M S **C** W	3	3	3
Comments / writeback in the visual Planning, annotations	M S C **W**	5	2	1
Maintainable by the business department Can be adjusted without coding skills	M **S** C W	5	5	2
Copilot / AI in the visual Copilot only creates and edits reports with core visuals	M S **C** W	1	1	1
AI-assisted authoring via PBIR / agent skills Agents write report JSON (PBIR); Vega-Lite specs and DAX UDFs are text	M S **C** W	2	3	3
Accessibility (keyboard, screen reader, contrast) WCAG 2.1 / EN 301 549; mandatory for public bodies (BITV 2.0), for B2C offerings BFSG since 28	M S **C** W	3	3	2

REQUIREMENT	PRIORITY	PAID 3rd-party paid	FLEXIBLE · OFF THE SHELF Open source / in-house	FLEXIBLE · CODE Deneb / Vega
June 2025 — the BFSG does not apply to purely internal reporting				
Control over the visual version Freeze a version, test it, schedule the rollout yourself	M **S** C W	2	5	3
Travels with Git and the deployment pipeline Promote the standard from dev through test to prod without manual work	M S **C** W	2	2	4
Standard centrally controllable via a theme file Colours, fonts and spacing once in theme.json, rolled out across all reports	M **S** C W	3	3	3
Mobile use (Power BI Mobile app) Rendering and operating on phone and tablet, dedicated mobile layout	M S **C** W	4	3	2
Copying into PowerPoint and Office “Copy visual” as an image with a data link into slides and documents	M S **C** W	2	2	3
Paginated reports (RDL)	M S **C** W	1	1	1

REQUIREMENT	PRIORITY	PAID 3rd-party paid	FLEXIBLE · OFF THE SHELF Open source / in-house	FLEXIBLE · CODE Deneb / Vega
Printed report booklet, pixel-perfect pages, batch output				
Ability to exit / no vendor lock-in Source code access, portability of the definitions	M S **C** W	1	3	4
Consistent design without styleguide discipline Recognisable out of the box	M **S** C W	5	4	2
Runs in Report Server (on-prem) Licensed visuals are not supported there	M S C **W**	1	2	2
Consistent scaling and zero baseline IBCS rule: the same scales across all charts on a page	M **S** C W	5	4	4
Change history / code review of the visual definition Git diff, revision, traceability	M S **C** W	2	3	5
Contractual support with a response time SLA, escalation path, named contact	M **S** C W	5	1	2
No data processing outside the tenant	M **S** C W	4	3	5

REQUIREMENT	PRIORITY	PAID 3rd-party paid	FLEXIBLE · OFF THE SHELF Open source / in-house	FLEXIBLE · CODE Deneb / Vega
The visual calls no external services				
Knowledge held by more than one person Changes possible without a key person	M S C W	5	3	2

1 = not possible · 2 = workaround only · 3 = limited · 4 = good · 5 = very good

Default: Daten-WG assessment, as of 09/2026. Your own ratings have a dotted border and are included in the share link.

LEVEL 2

Red flags and governance switches

Hard constraints from the tenant, IT security or procurement. They act as knock-out filters before anything is calculated. Excluded options stay visible in the result, below the line and with a reason.

Distribution channel for the visual

open | AppSource | Org store | File import

No factor. In reality the distribution channel is the lever you pull first - deliberately neutral here as long as it is undecided.

- [ ] 🚩 Tenant allows certified visuals only
  "Add and use certified visuals only" is active. The tenant setting applies to visuals from AppSource, not to visuals from the Organizational Store: whatever the admins release there also loads uncertified. Without that path, non-certified visuals do not load in the Service.
- [ ] Our open-source visual is Microsoft-certified
  Removes the knock-out from "certified visuals only" for open source
- [ ] The OSS visual is provided through the Organizational Store
  Admin portal → organisational visuals; removes "certified visuals only", including for uncertified .pbiviz files and likewise for a Deneb standalone build. Does not reduce the approval hours (the security review stays with you); the path does not apply in Report Server.
- [ ] No external service provider without a tender
  Procurement law / purchasing policy: the external share is set to 0
- [ ] 🚩 Core visuals only
  Custom visuals blocked by tenant setting
- [ ] 🚩 An IBCS certificate is mandatory
  Purchasing or a group standard requires certified software
- [ ] 🚩 No external licence budget
  €0 for visual licences, internal time only. For a pilot and evaluation there are free paths: graphomate charts for Power BI is free and fully functional (unrestricted in Desktop, watermarked in the Service), Inforiver Premium Matrix is free on the desktop and only becomes licensable when you publish. That is not enough for a viewer-wide standard, so the knock-out stands.
- [ ] Reports also run in Report Server
  On-premises
- [ ] Publish to web or Embedded (app owns data)
  Licence enforcement for paid visuals is not possible there
- [ ] Sovereign Cloud / GCC
  Restricted AppSource access
- [ ] Certified-only in the Service only, no desktop group policy
  A common gap
- [ ] Procurement under German public procurement law / EU data residency required
  Public sector, utility contracting entities or a group policy: thresholds, tendering, data processing agreements and data residency have to be settled before the purchase. A warning, not a knock-out — the calculator cannot know which vendor meets the requirements.

- [ ] **Co-determination: the works council has to be involved**
  § 87 Abs. 1 Nr. 6 BetrVG (German Works Constitution Act) covers technical devices intended to monitor behaviour or performance; case law reads “intended” broadly. User-level licence checks and product telemetry can fall under it. A warning, not legal advice — clarify the wording and its application with your own legal department.
- [ ] **Internal audit demands insight into the calculation logic**
  Internal audit, external auditors or a supervisory function want to trace how a deviation, a subtotal or a variance figure in the chart comes about.
- [ ] **No full input VAT deduction**
  Banks, insurers, hospitals, the public sector: VAT becomes a real cost. Otherwise the calculator works net. Increases all cash outflows (licences, support, capacity, courses, external hours) by the non-deductible share — the rate is in step 4 under “Non-deductible input VAT”. Hits paid hardest, because the cash share is largest there.
- [ ] **Budget approval only in the following year**
  The request is in progress, the money arrives in the next financial year: licence, setup, training and rollout start a year later, the horizon stays the same. A net present value advantage for the licence paths and a disadvantage for the hours paths, because the benefit starts later.
- [ ] **Capacity is already running at its limit (overage)**
  Consumption above the booked SKU is billed as overage at three times the pay-as-you-go rate (Microsoft Learn, preview). That is the real marginal price; without the switch the capacity uplift is calculated linearly. Only takes effect if annual capacity costs > 0 are entered.

LEVEL 2

Result and assessment

Total cost over the horizon, cost per viewer per year, upfront investment against running cost. The ranking is an indication under uncertainty in the inputs, not a price quote. Alongside P50 the cards also give the **net present value (NPV)** — later payments discounted at the discount rate; over three years that barely makes a difference, over ten it does. The reading guide below says in four steps what is where.

Reading guide: four blocks, in this order

CLICK TO JUMP THERE

If you are seeing the result for the first time, read it from top to bottom in this order. Clicking a row jumps to that block and highlights it briefly.

1. **Decision path — why this result?**
   The questions in the order in which they turn the result. What to watch: the provenance colour of each node — evidenced, rule of thumb or opinion.

2. **Ranking with bands — how solid is the gap?**
   One card per approach with P50 (the middle number) and the P10–P90 band. What to watch: if two bands overlap, the order does not hold — then the requirements decide, not the price.

3. **Report classes — what mandatory IBCS costs**
   How many reports actually need an IBCS chart. What to watch: what the “IBCS mandatory for all reports” switch does in your case — since v0.14 it works in both directions: paid, open source and Deneb get cheaper (the operational class A reports move out of the second toolchain and into the selected approach), core + SVG gets more expensive (there only the conformance review is left). The decision path works out both directions with your numbers.

4. **Evidence bar — what the number rests on**
   Splits the cost per approach by where the driving number comes from. What to watch: a high share of estimates does not mean “more expensive”, it means “less well evidenced”.

Key terms explained

P10 / P50 / P90
: In 10%, 50% and 90% of the 1,500 simulation runs the cost was below this value. P50 is the “middle” number, P10 to P90 the range you should plan with.

PERT
: A three-point estimate (min, likely, max) from which a probability distribution is formed. The expected value is (min + 4·likely + max) / 6.

MoSCoW
: Must (cannot do without it), should (important), could (nice to have), won't (not this time). A must with fulfilment level 1 is a knock-out.

Knock-out
: The approach is excluded but stays visible with its cost, so that the price of the constraint can be seen.

net present value (NPV)
: Later payments discounted at the discount rate. Over 10 years it makes a difference, over 3 hardly any.

Volume tiering
: Volume discount on the licence: the more users, the cheaper the individual seat. A negotiated vendor quote is calculated without tiering, because the discount is already in it.

Red flag
: A hard constraint from the tenant, IT security or procurement that rules out an approach before anything is calculated.

Price anchor
: The assumed list price per **licensed** user per year, as long as there is no real quote. The licensed headcount is all report authors plus the share of viewers with access to standard reports, not all viewers. It flips the ranking more than any other assumption.

Decision path

SEVEN NODES · DECISION MEASURE P50

The order in which this decision is really made. Each node shows what it screens out or pushes to the front in the current scenario, and how well evidenced it is. Dashed chips drop out, filled ones lead.

1 Constraints

No red flag active. All four approaches stay in the running.

`all four still open`

`evidenced`

2 Must requirements

1 criterion set as must (IBCS-compliant notation). No approach fails on a must. Only via a workaround, or not assessed: Core + SVG (IBCS-compliant notation).

`Core + SVG`

`opinion`

3 Report authors : viewers

Currently **1 : 30**. Tipping point at about **176 viewers**, and you are below it: the licence weighs less than the extra hours. Chart mix 44 / 39 / 17%.

`Paid`

`evidenced`

4 Report classes and mandatory IBCS

80% of the reports need a special IBCS graphic at all (40% standard, 40% demanding). The remaining **20%** are operational lists and tables: they are built with the chosen approach only because of the mandate. The “IBCS for all reports” mandate is switched on and costs, over 5 years: Open source **−€1,100** · Deneb **−€1,100** · Core + SVG **+€2,000** · Paid **+€2,700**. That **always** contains the review process (for Paid +€2,100, equally expensive in every approach, because a conformance review is a human pass); the rest is the shift of the operational reports into the tool, which saves a second toolchain and can therefore take cost out. With core + SVG exactly the review process is what remains.

`Open source` `Paid`

`opinion`

5 Price anchor and library

Without the library: Inforiver €33 → **Paid**, blended price €80 → **Paid**, Zebra €170 → **Open source**. With the library: Inforiver €33 → **Paid**, blended price €80 → **Deneb**, Zebra €170 → **Deneb**. The anchor turns the result. Currently: blended price €80, library off.

`Paid` `Open source` `Deneb`

`rule of thumb`

6 **Operating model**

Managed Self-Service, self-service share 25%: 2 of 5 report authors build templates. Variants ×1.5, governance ×1.15, support ×0.93, reuse ×0.95. Hit hardest: **Core + SVG** (variant effect 1.42× on maintenance and approval; maintenance and governance are 48% of its cost).

Core + SVG

opinion

7 **Result**

Rank 1–2 **Paid** €130,000 · Rank 1–2 **Open source** €140,000 · Rank 3 **Deneb** €160,000 · Rank 4 **Core + SVG** €240,000 (P50). The weighted rank (70% cost) confirms rank 1. 13% of the Paid cost rests on evidenced prices, 43% on estimates. Payback against the status quo in year 1, without reader time in year 2.

Paid

rule of thumb

evidenced = price list, snippet or documented product limit

rule of thumb = literature without measurement

opinion = panel or authors' estimate

3rd-party paid (Paid) is, at **€130,000** (P50, middle case) over 5 years, the best-placed permissible approach (P10–P90: €100,000 to €180,000), **€144** per viewer per year. Requirements score 72%. Ranks 1 and 2 cannot be told apart statistically, their bands overlap. Uncertainty: P90 is **1.8×** P10. Ranking: 70% cost, 30% requirements · operating model Managed Self-Service (2 of 5 report authors build templates).

These ranges come out of the chosen model assumptions. They are not an empirically validated forecast of actual project cost. The ranking holds only for the scope described (standard charts, no data model, no data preparation), the product families considered and the constraints shown. If you change an assumption, the badge still shows where the original value came from.

Ranking: cost ⟷ requirements met

70% cost · 30% requirements

100% = ranking by cost alone. Move it to the left to give weight to soft benefits (should/could fulfilment) as well. Weighted-scoring caveat: weights are subjective, which is why they are visible and adjustable.

Decision measure

P50 middle case	P75 cautious	P90 near worst case

Ranking, headline, key figures and the overview sentence all use this quantile. **Ranking by P90 = conservative:** the approach with the wider distribution loses its lead. As measured, the order only flips in the corporate preset (as of v0.19: by P50 and P75 Deneb leads, by P90 paid does — Deneb's wider distribution costs it the lead there); in the other four presets only the gap changes.

Draw assumptions paired across approaches

paired	independent

Paired: an optimistic hours assumption applies to all four approaches at once in each scenario, so differences are comparable pairwise. Independent: every approach draws its own random values, and then part of the difference comes from the sampling.

PAID

Rank 1–2

3rd-party paid

€130,000 5 yrs

P50, middle case · P10–P90 €100,000 – €180,000 · NPV €120,000

€144 per viewer per year

Upfront €22,000 · running €22,000/year
First standard report ≈ 78 h of working time · at the earliest after ≈ 12 weeks · each further one in the final year ≈ 5.6 h (marginal effort)
681 h internal over 5 years · 257 h in the first year = 0.2 FTE (1,500 h per FTE and year)
Exit after 5 years: 475 h ≈ €53,000 (lock-in 93%, not part of the total)

Requirements score 72%

FLEXIBLE · OFF THE SHELF

Rank 1–2

Open source / in-house

€140,000 5 yrs

P50, middle case · P10–P90 €88,000 – €230,000 · NPV €130,000

€148 per viewer per year

Upfront €37,000 · running €20,000/year
First standard report ≈ 91 h of working time · at the earliest after ≈ 7 weeks · each further one in the final year ≈ 9.6 h (marginal effort)
1,209 h internal over 5 years · 445 h in the first year = 0.3 FTE (1,500 h per FTE and year)
Exit after 5 years: 782 h ≈ €87,000 (lock-in 79%, not part of the total)

Requirements score 51%

⚠ Should “Contractual support with a response time”: No SLA; consulting on request, otherwise community support

FLEXIBLE · CODE

Rank 3

Deneb / Vega

€160,000 5 yrs

P50, middle case · P10–P90 €100,000 – €260,000 · NPV €150,000

€176 per viewer per year

Upfront €49,000 · running €23,000/year
First standard report ≈ 112 h of working time · at the earliest after ≈ 4 weeks · each further one in the final year ≈ 10.3 h (marginal effort)
1,194 h internal over 5 years · 464 h in the first year = 0.3 FTE (1,500 h per FTE and year)
Exit after 5 years: 503 h ≈ €59,000 (lock-in 40%, not part of the total)

Requirements score 48%

MACGYVER

Rank 4

Core visuals + SVG

€240,000 5 yrs

P50, middle case · P10–P90 €150,000 – €410,000 · NPV €230,000

€259 per viewer per year

Upfront €67,000 · running €35,000/year
First standard report ≈ 151 h of working time · at the earliest after ≈ 1 week · each further one in the final year ≈ 16.6 h (marginal effort)
1,856 h internal over 5 years · 684 h in the first year = 0.5 FTE (1,500 h per FTE and year)
Exit after 5 years: 1,012 h ≈ €110,000 (lock-in 50%, not part of the total)

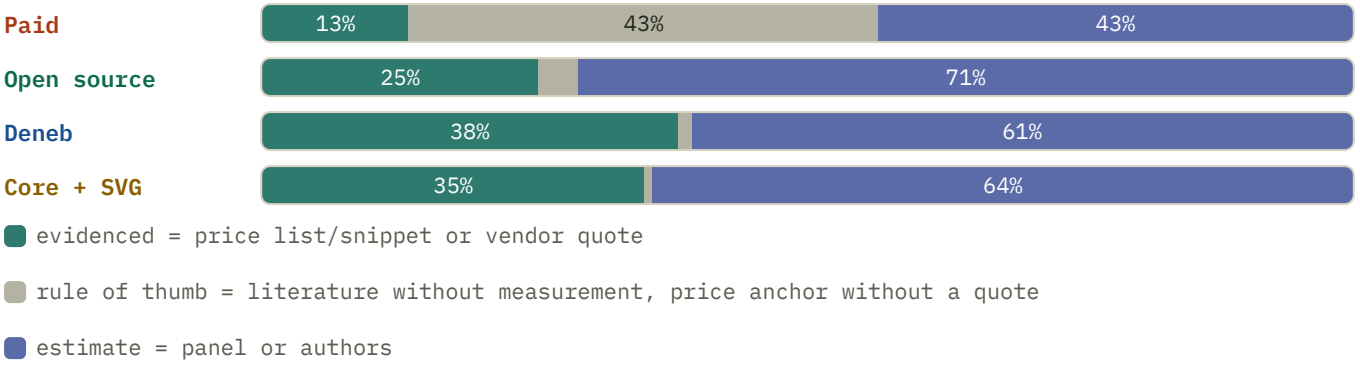
Requirements score 64%

⚠ Must “IBCS-compliant notation” only via a workaround: Only via workarounds (overlapping bars, SVG measures)
⚠ Should “Control over the visual version”: Core visuals follow the Power BI release train, freezing is not possible

How certain are these costs?

PROVENANCE OF THE DRIVING ASSUMPTIONS · SHARES OF THE P50 COST

The bar splits the cost not by item but by **where the number that drives it comes from**. An approach whose cost rests two thirds on estimates is not more expensive than one with evidenced prices — it is less well evidenced. Each cost category is assigned via its dominant parameter; the externally contracted share of the hours items counts as evidenced, because it runs on the market price for external hours. **Since v0.15 the licence price only counts as evidenced if a vendor quote (level 2) is active** — the three price anchors are search snippets: "there was a price somewhere" is no evidence for *your* price.



Report classes: what mandatory IBCS costs

EXPECTED VALUES OVER 5 YEARS · SHARES OF EACH APPROACH

Some of the reports need no special IBCS graphic at all. These reports (class A) are built with core visuals and cost the same in **every** approach — they dilute the comparison as long as they are counted in. The row "approach-independent" shows how much that is: initial build of the core chart types, rollout, maintenance and migration of those reports. They still drive licence, governance and user support, but those do not appear in this row. Across the approaches the row is **similar, but not identical**: the hours are the same, yet they are valued at the approach's blended hourly rate, and the estate is maintained at its maintenance rate — precisely those two figures stay attached to the approach even for operational reports. Without the mandate the conformance review applies only to classes B and C, with it to every report — **equally expensive in every approach**. That is the driver that has nothing to do with the tool.

ITEM	PAID
Total cost (expected value)	**€13(**
of which approach-independent: reports with no IBCS need (does not apply, IBCS is mandatory)	0% of the
of which IBCS conformance review (100% of the reports, plus a sample)	€1(8% of the
Remainder: what really depends on the chosen approach	€12(
Counter-calculation without the IBCS mandate	€13(
Price of the IBCS mandate broken down in the next two rows; all amounts rounded to two significant figures	**+€2**
of which the review process: conformance review for every report instead of only B and C	+€2
of which the shift: operational reports are built with the approach instead of with core visuals	+

How to read this: 20% of the reports are operational (class A), 80% need an IBCS graphic. The mandate has two effects, and they point in opposite directions. **The review process costs something in every approach** — for Paid an extra €2,100, and it depends on the procedure, not on the tool. **The shift**, on the other hand, can take cost out: if you build the operational reports with the purchased or self-built visual anyway, you save the second toolchain (a core construction kit of your own and its maintenance) — which is why the figures read Paid +€641, Open source −€3,100, Deneb −€3,100, Core + SVG +€0. In total the mandate costs least with **Open source** (−€1,100) and most with **Paid** (+€2,700). With core + SVG the shift is zero by definition, so exactly the review process is left over — the most honest view of what the rule costs. Both assumptions behind it (hours per review, annual sample) are authors' estimates with no source, see level 3.

Do-nothing option: no standard

THE STATUS QUO AS A BENCHMARK, NOT PART OF THE RANKING

STATUS QUO OVER 5 YEARS

€300,000

P10–P90 €200,000 – €440,000 · €318 per viewer and year

OF WHICH READER TIME

€180,000

without reader time €130,000 · an assumption with no source

AD-HOC BUILDING, MAINTENANCE, FOLLOW-UP QUERIES

€140,000

€58,000 building · €68,000 maintenance · €10,000 follow-up queries

Paid (€130,000) comes out below the status quo (€300,000) **only if reader time counts**: from 0 minutes per viewer and week onwards (your assumption: 2). Without reader time the status quo is €3,200 cheaper. Those minutes are the least evidenced value in the whole calculator.

Payback against the status quo (cumulative cost, expected values): **Paid** pays back against the status quo in year 1, without reader time in year 2 · **Open source** pays back against the status quo in year 1, without reader time never within the horizon · **Deneb** pays back against the status quo in year 2, without reader time never within the horizon · **Core + SVG** pays back against the status quo in year 4, without reader time never within the horizon. If reader time does not count for you, the second figure is the decisive one — reader time is the most weakly evidenced item in the model.

The do-nothing option builds the same reports with core visuals and no template. It cannot deliver complex charts (a waterfall with scenario notation); that loss of value carries no price here. Assumptions under step 4, prefix “do-nothing option”.

Cash out per year

CASH = CASH OUTFLOW (LICENCE, SUPPORT, CAPACITY, COURSES, EXTERNAL HOURS) · INTERNAL TIME = INTERNALLY VALUED TIME

Expected values per approach, calculated year by year. Year 1 carries the build-out (setup, development, initial training, approval) and the first rollout cohort; after that the estate keeps growing while licence, maintenance, governance and support run alongside. €400k of licence invoices is a different request to sign off than €400k of time from a team that is on the payroll anyway.

YEAR	PAID			OPEN SOURCE		
	CASH	INTERNAL	TOTAL	CASH	INTERNAL	TOTAL
Year 1 (build-out + operation)	€16,000	€€22,000	€€37,000	€13,000	€€38,000	€€51,000
Year 2	€14,000	€€15,000	€€29,000	€9,700	€€26,000	€€35,000
Year 3	€12,000	€€7,300	€€20,000	€4,200	€€14,000	€€18,000
Year 4	€14,000	€€7,800	€€22,000	€4,300	€€14,000	€€19,000
Year 5	€17,000	€€8,300	€€25,000	€4,300	€€15,000	€€19,000
Total over 5 years	**€74,000**	**€€60,000**	**€€130,000**	**€35,000**	**€€110,000**	**€€140,000**

Capex, opex and the P&L impact

CAPITALISING INTERNALLY GENERATED DEVELOPMENT WORK · NOT ACCOUNTING OR TAX ADVICE

☐ Capitalise internal development hours
Without the tick the table only shows the split. With it, the capitalisable part is amortised over the useful life and the P&L impact of the first year of operation is shown.

Useful life: 3 years | **5 years**

ITEM
Capitalisable: setup, in-house development, initial build of the chart types
Expensed immediately: licence, support, capacity, training, governance, maintenance, risk, migration, rollout
Capitalisable share

The legal position: licences can never be capitalised — they are rights of use for a limited time. Internally generated intangible fixed assets are **an accounting policy choice under § 248 Abs. 2 HGB** (German Commercial Code: recognition permitted, not required), whereas under **IAS 38** they are **mandatory** once the criteria for the development phase are met; **§ 5 Abs. 2 EStG** forbids recognition in the tax balance sheet, so capitalisation takes effect under commercial law only. **Mandatory note:** capitalisation requires project-based time recording (no time records, no recognition) and has to be agreed with the auditor. Research-like preliminary work — evaluation, selection, feasibility — is an expense under IAS 38 as well; the calculator does not separate it, "setup" contains both. This is an orientation, not accounting or tax advice.

LEVEL 2 What this calculator cannot do

So that the numbers above are read correctly: **where the assumptions come from, which causal chains the model does not know, and in which direction it gets things wrong.**

What these numbers are

Indications under uncertainty. **Not a quote, not a business case, not a price statement.** The one exception: numbers from a vendor quote that you entered yourself in level 2 — those are as solid as your quote.

Where the assumptions come from

58 assumptions. 12 have an external source (3 verified, 9 from a search snippet whose vendor page we were unable to retrieve). **46 have none**: 3 rules of thumb, 9 from a simulated expert panel, 34 estimates by the authors. Every assumption carries its provenance as a badge in level 3.

The assumptions the ranking hangs on — **hours per chart, per approach** — are among the unevidenced ones. We found no solid literature for them; the values come from a simulated expert panel and from our own experience. The evidence bar shows, per approach, how much of the cost depends on them.

Disclosure

ChartKitchen byDatenWG (the example for "open source / in-house") and the Deneb templates are projects run by the organisers. So we are comparing our own work with purchased products here too.

Six known gaps

FROM AN INDEPENDENT REVIEW

An independent review of the model (09/2026) found seven causal chains that shift the result noticeably and that the calculator either does not know or gets wrong. One of them has been fixed: since v0.18 the licence volume knows about the report classes. Six are still open.

Gap	Magnitude	Works in favour of
The number and ambition of reports are price-inelastic — in reality, expensive approaches build fewer and simpler ones	10–40% of the volume	3rd-party paid
The rollout has no speed limit: the calculator warns about missing capacity but does not price it	delays payback by years	Deneb, core
No key-person risk: staff turnover is linear and identical for every approach	5–15% in small teams	Deneb, open source
Price escalation without bargaining power: a smooth annual rate instead of a jump when the contract rolls over	tail risk, hits P90	3rd-party paid
What has been learned does not decay along with staff turnover	10–20% of the rollout hours	Deneb, core
Extra capacity consumption does not grow with the existing estate and the number of viewers	2–5× where a capacity is entered	Core + SVG

Four of the six work in favour of the approaches we stand for ourselves. So the model is not neutral; it has a known bias in our own direction. The full review report, including suggestions for modelling these gaps, is openly available in the repository.

What is not included

- **Power BI base licences** (Pro, PPU, the base price of a capacity) — they apply equally to all four approaches.
- **Paginated reports (RDL)** — Power BI custom visuals do not exist there at all, neither purchased ones nor Deneb. A printed report pack gets built a second time in every approach.
- **The benefit side** — only as a requirements score and as "€ per requirements point", not as a return.
- **Tax questions** (e.g. German withholding tax under § 50a EStG on licences from foreign vendors). Capitalisation and amortisation are included as orientation, not as accounting or tax advice.

And the most important point

The tipping point — the core number of this calculator — has **moved by more than 30% twice**since v0.13, both times through a correction to the model, not through new data. The **direction** of the statements is robust; the **exact number** is not. If you use it as a decision threshold, recompute it with your own numbers instead of copying it from here.

METHOD, LIMITS, DATA PROTECTION

Method: For each approach the calculator adds up licence and support (licensed headcount × price × years; licensed headcount = all report authors plus the share of viewers with access to standard reports), build-out (setup, optional in-house development, chart types × hours × blended hourly rate), rollout (reports × charts × reuse hours), training (report authors × hours and course fees, recurring through staff turnover), licence with viewer growth and price escalation per year, maintenance (rate × build-out + rollout × years plus breaking updates) and discontinuation risk (probability × migration share × build-out). Every assumption is a three-point estimate from which 1,500 scenarios are drawn via a PERT distribution; P10, P50 and P90 are shown. Break-even and sensitivity use the PERT expected value. Knock-out criteria (a must with 0 points, governance switches) filter before the ranking; excluded approaches stay visible for orientation. The requirements score is a weighted scoring (must 3, should 2, could 1 as weights, fulfilment level 1 to 5) and deliberately enters the ranking only through the slider. Governance and user support grow with the number of viewers (log-linearly from 50 to 5,000 viewers); the operating model separates deeply trained template builders from template users and scales variants, support and governance. Licences grow with the licensed headcount and the price, hourly rates with wage inflation, the running internal hours fall with the productivity gain, and maintenance follows the trend by cohort age. The discontinuation risk is a hazard rate per year: in the expected value the sum over the years, in the simulation a drawn event year, with the damage always measured against that year's estate. The net present value applies the discount rate to the cash flow of each year. The ranking uses the cost ratio to the cheapest option (excluded ones included), so that a hopeless third option does not shift the order of the others; ranks with overlapping P10–P90 bands are merged. Amounts are rounded to two significant figures. Audit 09/2026 (three Opus and three Sonnet reviews, blind estimate, web verification): turnover training differentiated by role, discontinuation risk including 60% of the rollout, approval per visual package instead of per chart type, licence administration degressive and zero for a site licence, tiering per year, governance log-linear, valuation of internal hours as a constant; 22 defaults reset, provenance per value in the badge. **Every change since v0.8 is listed in the change history below.**

Change history v0.8 to today (15 entries)

- **v0.8** — chart mix by complexity (factors per approach, the panel hours apply to 50 / 35 / 15), Wright learning curve on reuse (panel value = the 10th chart per report author, floor 50%), the do-nothing option "no standard" as a benchmark outside the ranking, with reader time reported separately.
- **v0.10** — susceptibility to variants as an assumption of its own per approach instead of being tied to the design score, a "template library available" switch for Deneb and core (off by default, a project in its own right), price anchors on level 1 showing who leads under all three anchors. v0.11 (assumption audit round 2, package A): paired draws — one random stream per parameter, shared by all four approaches, switchable; rollout carries wage inflation on the growth arising over the horizon; cash out per year split into cash outflow and internal time (table and Excel sheet); payback against the do-nothing option with and without reader time; internal hours and FTE per approach with a capacity warning (1,500 h per FTE and year); decision measure P50 / P75 / P90

switchable (ranking by P90 = conservative); exit cost as a separate figure outside the total; a viewer cap against unlimited growth; the do-nothing option using the core complexity factors of the chart mix; the 50% floor applied to the product of library × learning curve instead of to the learning curve alone; new defaults with sources for hourly rates, wage inflation, licence price escalation, the Zebra anchor (derived), site licence, discontinuation risk for core and running governance for Deneb (2.0 on 8 September 2026).

- **v0.11 (package B)** — the red flag "certified visuals only" made precise (the tenant setting applies to AppSource, not to the Organizational Store) and a modifier "open-source visual via the Organizational Store" added, which lifts the knock-out without reducing the approval hours; three new requirements (AI-assisted authoring via PBIR / agent skills, accessibility, data export from the visual) and a corrected criterion "large data volumes", because the 30,000-row window is the SDK limit of every custom visual; source and note texts put right (reader-time study blueforte / TU München 2019 instead of Zebra BI, a currency band instead of "$1 ≈ €1, conservatively", a price increase as a change of model, the free desktop routes, cx0 and the learning-curve anchor named).
- **v0.12 (package C1, calculation core)** — the core now calculates year by year instead of with products of averages — estate and rollout cohorts per year, maintenance only on the existing estate and ageing from its own year of creation, licence with tiering, minimum bundles and a viewer cap per year, wage inflation and productivity gain per year, the learning curve on the cumulative number of charts built per report author. This makes the net present value, the cash flow and the cumulative chart dependable for the first time (the nominal total changes by only −4.5 to +0.1%, because a rollout that was too cheap and maintenance that was too expensive used to cancel each other out). Also new: a rollout period for the existing estate (no longer everything in year 1), a productivity gain per approach as the counterpart to wage inflation, a maintenance trend by cohort age instead of decay by calendar year, discontinuation as a hazard rate with a drawn event year, a shared team factor per scenario across all four approaches (the level band gets wider, the difference band stays narrow), minimum bundles and a minimum contract term for paid, lock-in on exit per approach, aligned complexity factors, and the marginal effort of the final year instead of the average over the horizon for "each further report". v0.12 (package C2, requirements and outputs): seven new requirements (control over the visual version, travelling with Git and a deployment pipeline, theme controllability, drilldown/drillthrough/report-page tooltip as a row of its own beside interactivity, which is now narrowed to selection and cross-highlighting, mobile, copying to PowerPoint, paginated reports) — 27 criteria in all; six new red flags (German public procurement, co-determination under § 87 Abs. 1 Nr. 6 BetrVG and the internal audit's right to inspect the calculation logic as warnings; no full input VAT deduction, budget approval only in the following year, and capacity at its limit as modifiers with a cost effect in the calculation core); four new inputs (adoption rate per approach, with the non-compliant remainder billed at do-nothing-option rates, semantic models as a multiplier on setup and initial build, distribution channel AppSource / Organizational Store / file import with factors on approval, running governance, breaking updates and lead time, and the lead time to production release as a display without cost effect); four new outputs (capex/opex split with amortisation and P&L impact under § 248 Abs. 2 HGB, IAS 38 and § 5 Abs. 2 EStG; an hours anchor panel / blind estimate / conservative + 50% for the four unmeasured hours figures; switchable uncertainty in the volume assumptions of ±10 or ±20%, paired across all approaches; an evidence bar that splits the P50 cost per approach by the provenance of the driving assumptions into evidenced, rule of thumb and estimate). Updated: Fabric SKU presets F2/F8/F64. Details in the Markdown version. The Monte Carlo page (step 6) additionally draws the volume assumptions per scenario with the chosen variation band and evaluates the distribution, the certainty of the ranking, cash against internal time, building against using, rank correlations (Spearman) and a scale simulation at a fixed report authors : viewers ratio. Paginated reports (RDL) are not included: Power BI custom visuals do not exist there, neither purchased nor Deneb nor core — a printed report pack is built a second time under all four approaches. Power BI base licences (Pro, PPU, capacity base price) fall due equally for all approaches and are not included, nor is the benefit side (only as a score and as "€ per requirement point") or tax questions (e.g. § 50a EStG for licences from foreign vendors). Capitalisation and amortisation have been an output of their own since v0.12 — as an orientation, not as accounting or tax advice.
- **v0.13** — decision path, impact map, decision tree.
- **v0.14** — report classes — the reports are split into three classes (A operational with no IBCS need, B IBCS standard, C IBCS demanding), each with its own share, its own number of charts per report and its own chart mix. Charts per report and the global chart mix have been derived figures ever since; an expert override can still force a single global mix. Class A reports are built with core visuals and therefore cost the core hours in every approach (a proportionate share of the initial build of the chart types, rollout, maintenance, migration); they still drive licence, governance and user support, but not the approach-specific chart hours. The switch "IBCS mandatory for all reports" pulls class A into the chosen approach as well and makes the conformance review compulsory for every report; without it, only what belongs to classes B and C is reviewed. The review is a new global assumption

(hours per report and review, plus an annual sample from the estate), the same for all four approaches, because checking a report against the IBCS rules is a human pass and does not depend on what the chart was built with — both values are authors' estimates with no source. The do-nothing option knows the classes too (with no standard every report is produced ad hoc, and the class mixes now only set the complexity) and has no conformance review: with no standard there is nothing to review.

- **v0.15** — vendor quotes — up to three real quotes can be entered in level 2 (vendor, model per user / volume tiering / site flat fee, price, number of users, minimum term, discount, support, price escalation clause, note). Exactly one can be active; for paid it replaces the price anchor, sets the licence model, the price or site flat fee, the minimum contract term, the price escalation clause and support as point values, and switches the volume tiering off, because a quoted price has already been negotiated (tiering on top would be a second discount). No new calculation routes arise from it. From two quotes onwards a table compares total cost, cost per viewer and year, rank among the four approaches, and the negotiation ceiling against the cheapest hours-based approach. A quote is the hardest evidence in the model and gets a source badge of its own (A); in the evidence bar the licence price has since counted as evidenced only when a quote is active — the three price anchors are search snippets and, without a quote, run in the middle class, which lowers the evidenced share of paid without a quote and lifts it markedly with one.
- **v0.16** — guided start — a mode ahead of level 1 (seven questions: report authors and viewers or a preset, the number of standard reports and the share of reports needing IBCS via three report-class templates, operating model, five common red flags, price anchor or vendor quote, horizon, close) that only sets existing fields and clicks the existing controls; no new calculation logic, no changed assumptions, no interim state of its own — every answer lands in the inputs at once and the result recalculates live. Alongside it a reading guide in the result section of level 2 that jumps in four steps to the decision path, the ranking with its bands, the report classes and the evidence bar, plus simplifications on levels 1 and 2: technical terms (P50, PERT, MoSCoW, knock-out, volume tiering, net present value, red flag, price anchor) are explained in half a sentence where they first appear, the glossary "Key terms explained" has moved forward from level 3 to level 2, charts per report are rounded to one decimal place in running text, and four fine adjustments (semantic models, adoption rate, uncertainty in the volume assumptions, hours anchor) have moved from level 2 to level 3 — with a note at their old place, unchanged defaults and an unchanged calculation. A link with settings skips the guided start; the guided start itself is not stored in the link (sv stays 4) and does not appear in the printout.
- **v0.17** — table of contents as a sidebar that travels with you from level 2 onwards (all sections and the main blocks of the result, grouped by page area (by level up to v0.18), the active position tracked by IntersectionObserver; entries on deeper levels switch the level when clicked instead of greying out, and open closed details) with the buttons "Expand all" and "Collapse all" for reading in one go and for printing; a clickable impact map — selecting an input, an assumption group or a cost item highlights the connected paths, fades the rest back and explains underneath where the selection feeds in and how much it accounts for (amounts and shares from the same categories as before, provenance of the assumptions as in the evidence bar); the sensitivity line "all assumptions in this group 10% higher ≈ €x" uses expected(key, override), i.e. the route of the tornado chart, and is deliberately not shown for volume-assumption inputs, because there is no clean way to do it for them. On top of that the map becomes a second user interface: clicking a node opens a popup docked to that element — for inputs the matching existing control (number field, segment switch, report classes), for assumption groups the assumptions of that group with min · likely · max, source badge and short rationale as on level 3, for cost items only the composition; derived figures (charts per report, chart mix) stay unchangeable and point to the report classes. Every popup shows the effect as the result before and now, for the selected approach and for the cheapest permissible one, and has a "reset" button back to the state it had when it was opened. Only the existing fields and handlers are operated; no new calculation logic, no new state and no new share-link entry arise (sv stays 4). The map can also be reached directly from the table of contents, from the "Model as a map" button in the toolbar and from a third option at the end of the guided start.
- **v0.18** — licensed headcount separated from the volume assumptions — an independent review (causal-relationship check 09/2026, finding 1) established that although the hours of the report classes have been calculated separately since v0.14, the licence volume for 3rd-party paid still ran across all viewers, even though some of the readers never open a report built with the purchased visual. The new global assumption for this is "share of viewers with access to standard reports" (60 · 85 · 100%): only that share of the viewers plus all report authors make up the licensed headcount, and volume tiering and the minimum bundle apply to it as well; governance, user support, access governance and reader time stay on the full number of viewers, and the site licence is untouched. With the "IBCS mandatory for all reports" switch the share is necessarily 100% — the mandate lifts the licensed headcount back to all readers, and that is the second, so far invisible half of its price. The mechanism is evidenced by a search snippet (Zebra BI and Inforiver tie the licence scope to access to the reports built with the tool, not to the tenant); the level is an authors' estimate. **The correction works in favour of 3rd-party paid and against the approaches the authors run**

themselves: the licence block of paid, and with it the tipping point, shift by roughly the size of the viewer share taken out; the negotiation ceiling has since been labelled as a price per **licensed** user and year and sits correspondingly higher. Links from older versions (sv 4 and earlier) keep calculating unchanged, because they are set to 100% when loaded.

- **v0.19** — information architecture after Shneiderman — overview first, then details. Up to v0.18 the impact map was a collapsed details block on level 3 at the foot of the page; it now stands as an open section of its own, “What drives the cost”, on level 2 directly after the overview and before the inputs, because it is the overview of the model and not one of its details. The detailed reading instructions sit below it in a collapsed block; the decision tree without a scenario of its own stays a detail of level 3. Since then the table of contents groups by page area (guided start, overview and map, inputs, result, details) instead of by level, because the order of the page and the order of the levels are not the same. Calculation logic, assumptions and share link (sv stays 5) are unchanged.
- **v0.20** — Accessibility, states and keyboard; compliance check in the cash flow, VAT on course costs; print as a conference handout.
- **v0.21** — Debounced recalculation of the assumption fields; controls scoped to their own container (freeze in the map popup fixed).
- **v0.22** — One file, two languages: DE/EN switch top right, dictionary inside the document; expert-panel origin in the evidence bar and legend; long jumps snap instead of gliding for seconds.
- **v0.23** — Review round 3: tipping point and negotiation ceiling each come from one function, solved on the model instead of approximated (before: two values on one page); volume tiering as the cheapest covering tier, so Paid never gets cheaper with more viewers (where the next tier with its minimum quantity is cheaper, the Paid licence drops: enterprise preset −7%); the decision tree computes its ratios live; upper bounds for learning rate, external share and productivity gain; budget year fixed; open-source assumptions checked against manifest and docs (data processing and accessibility 3 each, texts and sources); five scenario presets. Share link sv stays 5.

All figures are indications under uncertainty, not quotes — with one exception: the figures you entered yourselves from a vendor quote. The calculation runs locally in the browser. The share link carries your inputs unencrypted in the URL fragment; whoever has the link sees the assumptions — **including the vendor quotes you entered, with vendor, price, terms and note, which also appear in the Excel sheet “Quotes”**. Fonts and the Excel library SheetJS are served by this website itself, not by third parties

`Markdown version (model, assumptions, sources)` · Visual Standards Calculator v0.23 · Michael Tenner & Diana Ackermann · Daten-WG Knowledge Kitchen · as of 11/2026 · Pre-Conference “Visual Standards in Power BI für Controlling und Finance”

Daten-WG · *Knowledge Kitchen*

Podcast, Tutorials, Werkzeuge und Datenstorys rund um Power BI, Microsoft Fabric und ehrliches Reporting.

FOLGEN

Alle Folgen
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Suche
Library

LERNEN

Dein erstes Dashboard
Power BI von A bis Z
Microsoft Fabric End-to-End
IBCS-Trainer
Fabric Alchemie
Daten-Imperium

WERKZEUGE

MockupKitchen
Chart-Builder
ChartKitchen Visual
P&L-Visual
Visual-Standards-Rechner
Alle Werkzeuge

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